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Motor vehicle

Abstract

A motor vehicle including a drive unit with an electric machine, wherein the electric machine is a directly cooled electric machine, a transmission, a first cooling loop for cooling assemblies of the directly cooled electric machine using a first dielectric coolant which flows directly around respective assemblies of the electric machine, a second cooling loop for cooling the transmission using a second coolant designed as a transmission oil, and a third cooling loop using a third coolant. The first cooling loop and the third cooling loop are coupled via a first heat exchanger, and the second cooling loop and the third cooling loop are coupled via a second heat exchanger.

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Background/Summary

CROSS-REFERENCE TO PRIOR APPLICATION

(1) This application claims benefit to German Patent Application No. DE 10 2022 103 357.0, filed on Feb. 14, 2022, which is hereby incorporated by reference herein.

FIELD

(2) The invention relates to a motor vehicle.

BACKGROUND

(3) Motor vehicles configured as electric vehicles and as hybrid vehicles are well known from practical experience. Both an electric vehicle and a hybrid vehicle have drive units comprising an electrical machine. The electric machine must be cooled during operation. Likewise, a transmission of the motor vehicle must be cooled during operation. Until now, it has been difficult to efficiently cool both the electric machine and the transmission of a motor vehicle and have a compact design.

(4) DE 10 2019 117 893 A1 discloses a motor vehicle, the drive unit of which comprises an electric machine. The motor vehicle comprises a cooling loop, wherein the coolant flowing through the cooling loop flows through both a transmission and power electronics as well as an electric machine. The coolant serves to cool both the rotor and the stator of the electric machine. This cooling loop is coupled to a further cooling loop via a heat exchanger, which serves to cool a battery.

(5) DE 10 2019 117 637 A1 discloses a further motor vehicle, the powertrain of which comprises an electric machine and a transmission. Two cooling loops are disclosed, wherein the first coolant of the first cooling loop flows through the transmission and the rotor of the electric machine, and the second coolant of the second cooling loop flows through the stator of the electric machine. Both cooling loops are coupled via a heat exchanger.

(6) US 2009/0 195 093 A1 and JP 2000 264 076 A disclose further prior art.

SUMMARY

(7) In an embodiment, the present disclosure provides a motor vehicle comprising a drive unit comprising an electric machine, wherein the electric machine is a directly cooled electric machine, a transmission, a first cooling loop for cooling assemblies of the directly cooled electric machine using a first dielectric coolant which flows directly around respective assemblies of the electric machine, a second cooling loop for cooling the transmission using a second coolant designed as a transmission oil, and a third cooling loop using a third coolant. The first cooling loop and the third cooling loop are coupled via a first heat exchanger, and the second cooling loop and the third cooling loop are coupled via a second heat exchanger.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached drawings, which illustrate the following:

(2) FIG. 1 shows a block diagram of a cooling system of a motor vehicle according to an embodiment the invention;

(3) FIG. 2 shows a block diagram of a cooling system of a motor vehicle according to an embodiment the invention; and

(4) FIG. 3 shows a block diagram of a cooling system of a motor vehicle according to the invention;

DETAILED DESCRIPTION

(5) In an embodiment, the present invention provides a motor vehicle having a compact and efficient cooling system.

(6) The motor vehicle according to an embodiment of the present invention has a drive unit comprising an electric machine, wherein the electric machine is a directly cooled electric machine. The motor vehicle according to an embodiment of the invention further comprises a transmission. The motor vehicle according to an embodiment of the invention further comprises a first cooling loop for cooling the directly cooled electric machine using a first, dielectric coolant, which flows directly around assemblies of the electric machine, and comprises a second cooling loop for cooling the transmission using a second coolant designed as a transmission oil, and comprises a third cooling loop using a third coolant, wherein the first cooling loop and the third cooling loop are coupled via a first heat exchanger, and the second cooling loop and the third cooling loop are coupled via a second heat exchanger.

(7) The motor vehicle according to an embodiment of the invention enables efficient cooling of at least the electric machine and the transmission along with a cooling system that is compact in design. The first coolant of the first cooling loop and the second coolant of the second cooling loop are not cooled directly against the environment, but rather against the third coolant of the third cooling loop. Doing so enables the cable lengths to be reduced and heat dissipation to be made more efficient.

(8) Preferably, the first heat exchanger and the second heat exchanger are connected in series to one another such that the third coolant first passes through the first heat exchanger and then through the second heat exchanger, or first passes through the second heat exchanger and then through the first heat exchanger. Alternatively, the first heat exchanger and the second heat exchanger are connected to one another in parallel such that a first portion of the third coolant passes through the first heat exchanger, and a second portion of the third coolant passes through the second heat exchanger.

(9) In order to ensure a design that is as compact as possible, the two heat exchangers of the first cooling loop and the second cooling loop are preferably connected to one another in a row, i.e., in series. It is in this case preferably provided that the third coolant first flows via the first heat exchanger of the first cooling loop and then via the second heat exchanger of the second cooling loop. In this case, the electric machine and the transmission are then cooled according to their thermal criticality.

(10) Preferably, the first cooling loop comprises a first feed pump, and the second cooling loop comprises a second feed pump, which pumps are designed as tandem pumps. The first feed pump conveys the first coolant at a first feed rate, and the second feed pump conveys the second coolant at a second feed rate. A tandem pump aids in the compact design of the cooling system.

(11) Preferably, the first feed rate is greater, particularly at least 50% greater, than the second feed rate. Effective cooling of both the electric machine and the transmission is possible via these feed rates.

(12) Preferably, the first cooling loop is configured such that the first coolant flows directly around at least windings of a stator of the electric machine. The second cooling loop is preferably configured such that the second coolant thereof also cools a shaft and/or windings of a rotor of the electric machine in addition to the transmission. Cooling the windings of the stator via the first cooling loop (and therefore the first coolant), and cooling the shaft and/or windings of the rotor via the second cooling loop (and therefore the second coolant) is preferred. Alternatively, however, it is also possible to cool the windings of the stator as well as the shaft and/or the windings of the rotor via the first cooling loop (and therefore the first coolant).

(13) Preferably, the third cooling loop comprises an air-water heat exchanger and is preferably configured such that the third coolant thereof cools a pulse inverter and/or a torque vectoring machine. Doing so enables advantageous cooling of the pulse inverter, as well as an optional torque vectoring machine, and a compact design of the cooling system.

(14) FIG. 1 is a block diagram of assemblies of an motor vehicle **10**, in particular of assemblies of a cooling system **11** of the motor vehicle **10**, which are used at least for cooling an electric machine **12** of a drive unit of the motor vehicle **10**, as well as for cooling a transmission **13** of the motor vehicle **10**.

(15) In an exclusively electrically driven motor vehicle, the motor vehicle **10** comprises only the electric machine **12**. An internal combustion engine is additionally present in a hybrid vehicle. The transmission **13** is connected on the drive side, between the drive unit and a power takeoff, wherein the transmission provides a tractive effort from the power unit to the power takeoff of the motor vehicle **10**.

(16) The motor vehicle **10** comprises a first cooling loop **14**, around which a first coolant for cooling the electric machine **12** flows.

(17) The electric machine **12** is in this case a directly cooled electric machine, wherein in order to cool the directly cooled electric machine, the first coolant, which is designed as a dielectric coolant, flows around the assemblies of said electric machine.

(18) The motor vehicle **10** further comprises a second cooling loop **15**, which serves to cool at least the transmission **13**. A second coolant flows around the second cooling loop **15**, wherein the second coolant is transmission oil.

(19) The motor vehicle further comprises a third cooling loop **16**. A third, in particular water-based, coolant flows around the third cooling loop **16**.

(20) The third cooling loop **16** is coupled to the first cooling loop **14** via a first heat exchanger **17**. In addition, the third cooling loop **16** is coupled to the second cooling loop **15** via a second heat exchanger **18**.

(21) Therefore, the first coolant of the first cooling loop **14** and the second coolant of the second cooling loop **15** are recooled via the third coolant of the third cooling loop **16** via the two heat exchangers **17**, **18**.

(22) In FIG. 1, the two heat exchangers, i.e., the first heat exchanger **17** of the first cooling loop **14** and the second heat exchanger **18** of the second cooling loop **15**, are connected to one another in series, i.e., such that the third coolant flowing through the third cooling loop **16** first flows through the first heat exchanger **17** and then through the second heat exchanger **18**.

(23) According to FIG. 2, the direction of flow of the third coolant through the third cooling loop **16** can also be reversed, wherein in FIG. 2 the third coolant of the third cooling loop **16** first flows through the second heat exchanger **18** of the second cooling loop **15** and only then flows through the first heat exchanger **17** of the first cooling loop **14** (i.e., when viewed in the direction of flow of the third coolant **16**, based on an air-water heat exchanger **19** of the third cooling loop **16**).

(24) The third, in particular water-based, coolant of the third cooling loop **16** as well as air flows

around the air-water heat exchanger **19** in order to cool the third coolant of the third cooling loop **16** via the ambient air.

(25) In FIG. **3**, the first heat exchanger **15** of the first cooling loop **14** and the second heat exchanger **18** of the second cooling loop **15** are connected to one another in parallel such that a first portion of the third coolant of the third cooling loop **16** then flows through the first heat exchanger **17**, and a second portion of the third coolant of the third cooling loop **16** flows through the second heat exchanger **18**. The heat exchangers **17**, **18** are preferably plate heat exchangers.

(26) The first cooling loop **14** comprises a first feed pump **20**, and the second cooling loop **15** comprises a second feed pump **21**. The two feed pumps **20**, **21** of the two cooling loops **14**, **15** are preferably designed as tandem pumps **22**.

(27) The first feed pump **20** of the first cooling loop **14** conveys the first coolant at a first feed rate. The second feed pump **21** of the second cooling loop **15** conveys the second coolant at a second feed rate.

(28) The first conveyance rate is in this case preferably greater than the second conveyance rate, in particular at least 50% greater.

(29) The first cooling loop **14**, which serves to cool the electric machine **12**, is in this case configured such that the first, dielectric coolant flows directly around at least windings of a stator **12a** of the electric machine **12**.

(30) The first coolant of the first cooling loop **14** then flows into a coolant reservoir **23** of the first cooling loop **14**. In order to cool and filter the first coolant of the first cooling loop **14** immediately upstream of the electric machine **12**, a filter **24** is connected between the first heat exchanger **17**, via which the first cooling loop **14** and the third cooling loop **16** are coupled, and the stator **12a** of the electric machine **12**.

(31) The filter **24** can be a fine filter, e.g., a molecular filter or a molecular sieve. The filter **24** ensures that no contaminants able to cause short circuits will enter the electric machine **12**.

(32) In the embodiment shown, the second cooling loop **15** serves not only to cool the transmission **13**, but also to cool a shaft as well as windings of a rotor **12b** of the electric machine **12**. When viewed in the direction of flow of the second coolant of the second cooling loop **15**, the rotor **12b** of the electric machine **12** is in this case connected to the transmission **13** in parallel.

(33) After flowing through the transmission **13** and the rotor **12b** of the electric machine **12**, the second coolant, which is designed as a transmission oil, enters an oil sump **25**, wherein in FIG. **1** a filter **26** is connected between the oil sump **25** and the second feed pump **21**, preferably immediately upstream of the feed pump **21**.

(34) In contrast to the preferential embodiment shown, it is also possible to directly cool both the rotor **12b** and the stator **12a** via the first coolant of the first cooling loop **14**.

(35) FIG. **1** illustrates a pulse inverter **27** of a power electronics means **28** as a further assembly of the motor vehicle, wherein the pulse inverter **27** of the power electronics means **28** is cooled by the third coolant of the third cooling loop **16**.

(36) If the motor vehicle comprises a torque vectoring machine **29**, then said machine can be cooled either via the third cooling loop **16** or via the second cooling loop **15**, as shown in FIG. **1**. If the torque vectoring machine **29** is cooled via the third cooling loop **16**, then said machine is preferably connected to the pulse inverter **27** in series. If the torque vectoring machine **29** is cooled via the second cooling loop **15**, then said machine is preferably connected to the transmission **13** in parallel.

(37) As previously stated, the second coolant in the second cooling loop **15** is used as a transmission oil.

(38) The first coolant of the first cooling loop **14** is a dielectric coolant, i.e., a dielectric oil.

(39) The third coolant of the third cooling loop **16** is in particular a water-based coolant, preferably a water-glycol mixture. A refrigerant or CO₂ may also be used as the third coolant of the third cooling loop **16**.

- (40) Preferably, the motor vehicle **10** is exclusively an electrically driven motor vehicle.
- (41) Embodiments of the invention enable efficient cooling of both the electric machine **12** and the transmission **13**, as well as the pulse inverter **27** and an optional torque vectoring machine **29** along with a compact design of the cooling system. The coolant used for cooling the electric machine and the coolant used for cooling the transmission are not cooled directly against the local environment, but rather against the third coolant of the third cooling loop. The direction of flow within the third cooling loop **16** is in particular matched to the thermal criticality of the components to be cooled. At least the windings of stator **12a** of electric machine **12** are directly cooled.
- (42) Preferably, the two feed pumps **20**, **21** of first cooling loop **14** and second cooling loop **15** are designed as tandem pumps.
- (43) The third cooling loop **16** preferably also comprises a feed pump.
- (44) While subject matter of the present disclosure has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. Any statement made herein characterizing the invention is also to be considered illustrative or exemplary and not restrictive as the invention is defined by the claims. It will be understood that changes and modifications may be made, by those of ordinary skill in the art, within the scope of the following claims, which may include any combination of features from different embodiments described above.
- (45) The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

Claims

1. A motor vehicle comprising: a drive unit comprising an electric machine, wherein the electric machine is a directly cooled electric machine; a transmission; a first cooling loop for cooling assemblies of the directly cooled electric machine using a first dielectric coolant which flows directly around respective assemblies of the electric machine; a second cooling loop for cooling the transmission using a second coolant designed as a transmission oil; and a third cooling loop using a third coolant, wherein the first cooling loop and the third cooling loop are coupled via a first heat exchanger, and the second cooling loop and the third cooling loop are coupled via a second heat exchanger.
2. The motor vehicle according to claim 1, wherein the first heat exchanger and the second heat exchanger are connected to one another in series such that the third coolant first flows through the first heat exchanger and then through the second heat exchanger, or first flows through the second heat exchanger and then through the first heat exchanger.
3. The motor vehicle according to claim 1, wherein the first heat exchanger and the second heat exchanger are connected to one another in parallel such that a first portion of the third coolant flows through the first heat exchanger, and a second portion of the third coolant flows through the second heat exchanger.
4. The motor vehicle according to claim 1, wherein the first cooling loop comprises a first feed

- pump, and the second cooling loop comprises a second feed pump, the first and second feed pumps being designed as tandem pumps.
5. The motor vehicle according to claim 4, wherein the first feed pump conveys the first coolant at a first feed rate, and the second feed pump conveys the second coolant at a second feed rate, wherein the first feed rate is greater than the second feed rate.
 6. The motor vehicle according to claim 1, wherein the first cooling loop is configured such that the first coolant flows directly around and cools at least windings of a stator of the electric machine.
 7. The motor vehicle according to claim 1, wherein the first cooling loop is configured such that the first coolant also flows directly around and cools a shaft and/or windings of a rotor of the electric machine.
 8. The motor vehicle according to claim 1, wherein the second cooling loop is configured such that the second coolant cools a shaft and/or windings of a rotor of the electric machine.
 9. The motor vehicle according to claim 1, wherein the second cooling loop is configured, and/or the third cooling loop is configured, such that the second coolant and/or the third coolant cools a torque vectoring machine.
 10. The motor vehicle according to claim 1, wherein the third cooling loop comprises an air-water heat exchanger.
 11. The motor vehicle according to claim 1, wherein the first cooling loop comprises a filter connected, in a direction of flow of the first coolant, between the first heat exchanger and the electric machine.
 12. The motor vehicle according to claim 1, wherein the first cooling loop comprises a coolant reservoir connected, in a direction of flow of the first coolant, between the electric machine and the first feed pump.
 13. The motor vehicle according to claim 1, wherein the second cooling loop comprises an oil sump and a filter, wherein the filter is connected, in a direction of flow of the second coolant, between the oil sump and the second feed pump.
 14. The motor vehicle according to claim 1, wherein the third coolant is a water-based coolant, or a refrigerant, or CO₂.
 15. The motor vehicle according to claim 1, wherein the motor vehicle is exclusively an electrically driven electric vehicle.
 16. The motor vehicle according to claim 5, wherein the first feed rate is 50% greater than the second feed rate.
 17. The motor vehicle according to claim 10, wherein the third coolant cools a pulse inverter.
 18. The motor vehicle according to claim 14, wherein the third coolant is a water-glycol mixture.
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